الخطوة 1: حصر الجداول والحقول — FirOb هيكل قاعدة بيانات

اصطلاحات: `ulid` = char(26) · `uBigInt` = unsignedBigInteger · `ts` = timestamp · `timestamps` = `created_at` + `updated_at` (ts, Nullable) · `SD` = SoftDeletes (`deleted_at` ts, Nullable). **قاعدة عامة:** `tenant_id` (ulid, FK → `tenants.id` , NOT NULL, INDEX) موجود في كل جدول مملوك لمستأجر، وكل `UNIQUE` يبدأ بـ `tenant_id` عليه مركّب . الحقول القابلة للترجمة (ar/en) من نوع `json` .

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الوحدة 1 — المستأجرون والباقات والفوترة

`tenants`

الحقل	النوع	القيود
id	ulid	PK
name	string(255)	NOT NULL
slug	string(100)	UNIQUE (عالمي)
domain	string(255)	Nullable, UNIQUE (عالمي)
status	enum(active, suspended, archived)	NOT NULL, Default `active`
logo_path	string(500)	Nullable
cached_user_count	unsignedInteger	Default 0
cached_storage_mb	uBigInt	Default 0

~~cached_storage_mb~~	~~uBigInt~~	~~Default 0~~
timestamps, deleted_at	ts	Nullable (SD)

`plans`

الحقل	النوع	القيود
id	uBigInt	PK, auto-increment
code	string(50)	UNIQUE (عالمي)
name	json	NOT NULL (ترجمة)
max_users	unsignedInteger	NOT NULL
max_storage_mb	uBigInt	NOT NULL
ai_token_quota_monthly	uBigInt	NOT NULL, Default 0
price_monthly	decimal(10,2)	NOT NULL, Default 0
is_active	boolean	Default true
timestamps	ts	Nullable

`subscriptions`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, ON DELETE CASCADE
plan_id	uBigInt	FK → plans.id, ON DELETE RESTRICT
status	enum(trialing, active, past_due, cancelled, expired)	NOT NULL
starts_at	ts	NOT NULL
ends_at	ts	Nullable
cancelled_at	ts	Nullable
timestamps	ts	Nullable
Index	`(tenant_id, status)`	

`tenant_settings`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE

key	string(150)	NOT NULL (مثل `security.require_2fa`)
value	json	Nullable
updated_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps	ts	Nullable
Unique	`(tenant_id, key)`	

`system_settings` **(Landlord، بدون tenant_id)**

الحقل	**النوع**	**القيود**
id	uBigInt	PK
key	string(150)	UNIQUE (عالمي)
value	json	Nullable
updated_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps	ts	Nullable

`invoices`

الحقل	**النوع**	**القيود**
id	uBigInt	PK
tenant_id	ulid	FK, RESTRICT
subscription_id	uBigInt	FK → subscriptions.id, RESTRICT
number	string(50)	NOT NULL
amount	decimal(10,2)	NOT NULL
currency	char(3)	NOT NULL, Default `USD`
status	enum(draft, open, paid, void, uncollectible)	NOT NULL, Default `draft`
issued_at	ts	NOT NULL
due_at	ts	Nullable
paid_at	ts	Nullable
timestamps	ts	Nullable
Unique	`(tenant_id, number)`	

`payments`

الحقل	**النوع**	**القيود**
id	uBigInt	PK

id	uBigInt	PK
tenant_id	ulid	FK, RESTRICT
invoice_id	uBigInt	FK → invoices.id, CASCADE
gateway	string(50)	NOT NULL
gateway_reference	string(150)	NOT NULL
amount	decimal(10,2)	NOT NULL
currency	char(3)	NOT NULL
status	enum(pending, succeeded, failed, refunded)	NOT NULL
payload	json	Nullable (بدون أرقام بطاقات)
paid_at	ts	Nullable
timestamps	ts	Nullable
Unique	`(tenant_id, gateway, gateway_reference)`	

الوحدة 2 — الهوية والمصادقة

`users`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, Nullable (NULL = Super Admin), CASCADE
name	string(255)	NOT NULL
email	string(255)	NOT NULL
email_verified_at	ts	Nullable
password	string(255)	NOT NULL
two_factor_secret	text	Nullable (مشفّر)
two_factor_recovery_codes	text	Nullable (مشفّر)
two_factor_confirmed_at	ts	Nullable
locale	string(5)	Default `ar`
remember_token	string(100)	Nullable
last_login_at	ts	Nullable
timestamps, deleted_at	ts	Nullable (SD)

Unique	`(tenant_id, email)`

`user_invitations`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
email	string(255)	NOT NULL, INDEX `(tenant_id, email)`
role_id	uBigInt	FK → roles.id, CASCADE
invited_by	uBigInt	FK → users.id, CASCADE
token_hash	string(64)	NOT NULL
expires_at	ts	NOT NULL
accepted_at	ts	Nullable
timestamps	ts	Nullable
Unique	`(tenant_id, token_hash)`	

`personal_access_tokens` **(Sanctum)**

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, Nullable, CASCADE
tokenable_type	string	NOT NULL (morph)
tokenable_id	uBigInt	NOT NULL (morph)
name	string	NOT NULL
token	string(64)	UNIQUE (عالمي)
abilities	text	Nullable
last_used_at	ts	Nullable
expires_at	ts	Nullable
timestamps	ts	Nullable
Index	`(tokenable_type, tokenable_id)`	

`password_reset_tokens`

الحقل	النوع	القيود

id	uBigInt	PK
tenant_id	ulid	FK, Nullable, CASCADE
email	string(255)	NOT NULL
token	string	NOT NULL (hash)
created_at	ts	Nullable
Unique	`(tenant_id, email)`	

`sessions`

الحقل	النوع	القيود
id	string	PK
user_id	uBigInt	FK, Nullable, INDEX
ip_address	string(45)	Nullable
user_agent	text	Nullable
payload	longText	NOT NULL
last_activity	integer	INDEX

الوحدة 3 — الأدوار والصلاحيات (Spatie Teams، `team_foreign_key = tenant_id`)

`roles`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, Nullable (team), CASCADE
name	string(125)	NOT NULL
guard_name	string(125)	NOT NULL
timestamps	ts	Nullable
Unique	`(tenant_id, name, guard_name)`	

`permissions`

الحقل	النوع	القيود
id	uBigInt	PK
name	string(125)	NOT NULL
guard_name	string(125)	NOT NULL

timestamps	ts	Nullable
Unique	`(name, guard_name)` (عالمي)	

`role_has_permissions`

الحقل	النوع	القيود
permission_id	uBigInt	PK, FK, CASCADE
role_id	uBigInt	PK, FK, CASCADE

`model_has_roles`

الحقل	النوع	القيود
tenant_id	ulid	PK, FK (team), CASCADE
role_id	uBigInt	PK, FK, CASCADE
model_type	string	PK
model_id	uBigInt	PK, INDEX `(model_id, model_type)`

`model_has_permissions`

الحقل	النوع	القيود
tenant_id	ulid	PK, FK (team), CASCADE
permission_id	uBigInt	PK, FK, CASCADE
model_type	string	PK
model_id	uBigInt	PK, INDEX `(model_id, model_type)`

الوحدة 4 — التدقيق والإشعارات

`activity_log`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, NOT NULL, RESTRICT
log_name	string	Nullable, INDEX
description	text	NOT NULL
subject_type / subject_id	string / uBigInt	Nullable (morph)

event	string	Nullable
causer_type / causer_id	string / uBigInt	Nullable (morph، بدون FK)
properties	json	Nullable (يحمل causer_id و tenant_id عند الطابور)
batch_uuid	uuid	Nullable
timestamps	ts	Nullable
Index	`(tenant_id, created_at)`, `(causer_id)`	

`notifications`

الحقل	النوع	القيود
id	uuid	PK
tenant_id	ulid	FK, Nullable, CASCADE
type	string	NOT NULL
notifiable_type / notifiable_id	string / uBigInt	NOT NULL (morph)
data	json	NOT NULL
read_at	ts	Nullable
timestamps	ts	Nullable

الوحدة 5 — المحتوى الأكاديمي

`subjects`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
code	string(50)	NOT NULL
name	json	NOT NULL
description	json	Nullable
is_active	boolean	Default true
created_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps, deleted_at	ts	Nullable (SD)
Unique	`(tenant_id, code)`	

Unique	`(tenant_id, code)`	

`concepts`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
subject_id	uBigInt	FK → subjects.id, CASCADE
code	string(50)	NOT NULL
title	json	NOT NULL
body	longText	Nullable (HTML معقّم بـ purifier)
status	enum(DRAFT, PENDING, PUBLISHED, REJECTED)	Default `DRAFT`
created_by / reviewed_by	uBigInt	FK → users.id, Nullable, SET NULL
submitted_at / reviewed_at / published_at	ts	Nullable
rejection_reason	text	Nullable
sort_order	unsignedInteger	Default 0
timestamps, deleted_at	ts	Nullable (SD)
Unique	`(tenant_id, code)` · **Index** `(tenant_id, subject_id, status)`	

`concept_media`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
concept_id	uBigInt	FK → concepts.id, CASCADE
mediable_type / mediable_id	string / uBigInt	Nullable (morph)
ai_generation_job_id	uBigInt	FK, Nullable, SET NULL
source	enum(manual, ai)	NOT NULL
resource_type	enum(text	NOT NULL

resource_type	enum(text, audio, image, pptx, pdf, docx, other)	NOT NULL
disk	string(20)	Default `r2`
path	string(500)	NOT NULL (`tenants/{tenant_id}/concepts/{year}/{month}/{hash}.{ext}`)
hash_name	string(100)	NOT NULL
mime_type	string(100)	NOT NULL
size_bytes	uBigInt	NOT NULL
visibility	enum(private, public)	Default `private`
metadata	json	Nullable
uploaded_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps, deleted_at	ts	Nullable (SD)
Unique	`(tenant_id, path)`	

`learning_objects`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
concept_id	uBigInt	FK → concepts.id, CASCADE
code	string(50)	NOT NULL
title	json	NOT NULL
type	enum(attachment, scorm, html5)	NOT NULL
disk	string(20)	Default `r2`
path	string(500)	NOT NULL
mime_type	string(100)	NOT NULL
size_bytes	uBigInt	NOT NULL
status	enum(DRAFT, PENDING, PUBLISHED, REJECTED)	Default `DRAFT`

is_public	boolean	Default false (للبوابة العامة)
created_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps, deleted_at	ts	Nullable (SD)
Unique	`(tenant_id, code)`	

الوحدة 6 — الذكاء الاصطناعي

`ai_generation_jobs`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
user_id	uBigInt	FK → users.id, CASCADE
concept_id	uBigInt	FK → concepts.id, CASCADE
resource_type	enum(text, audio, image, pptx, pdf, docx)	NOT NULL
provider	enum(openai, elevenlabs, replicate, openrouter, ollama, flux_local, piper_local)	NOT NULL
model	string(100)	Nullable
prompt	text	NOT NULL (معقّم)
params	json	Nullable
status	enum(queued, processing, completed, failed, cancelled)	Default `queued`
tokens_used	uBigInt	Default 0
error_message	text	Nullable
started_at / completed_at / cancelled_at	ts	Nullable
timestamps	ts	Nullable
Index	`(tenant_id, status)`	

`ai_usage_logs`

الحقل	النوع	القيود

id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
user_id	uBigInt	FK, Nullable, SET NULL
ai_generation_job_id	uBigInt	FK, Nullable, SET NULL
feature	enum(resource_generation, suggestion, question_generator, chatbot)	NOT NULL
provider	enum(نفس المزودين السبعة)	NOT NULL
tokens_used	uBigInt	NOT NULL
created_at	ts	NOT NULL
Index	`(tenant_id, created_at)`	

`chat_conversations`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
user_id	uBigInt	FK → users.id, CASCADE
title	string(255)	Nullable
provider	enum(نفس المزودين)	NOT NULL
model	string(100)	Nullable
timestamps, deleted_at	ts	Nullable (SD)
Index	`(tenant_id, user_id)`	

`chat_messages`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
conversation_id	uBigInt	FK → chat_conversations.id, CASCADE
role	enum(user, assistant, system)	NOT NULL
content	longText	NOT NULL (معقّم)

tokens_used	unsignedInteger	Default 0
created_at	ts	NOT NULL
Index	`(tenant_id, conversation_id, created_at)`	

الوحدة 7 — الاختبارات والتقييم

`quizzes`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
subject_id	uBigInt	FK → subjects.id, CASCADE
concept_id	uBigInt	FK → concepts.id, Nullable, SET NULL
title	json	NOT NULL
description	json	Nullable
time_limit_minutes	unsignedSmallInteger	Nullable
max_attempts	unsignedTinyInteger	Default 1
passing_score	decimal(5,2)	Default 50
shuffle_questions	boolean	Default false
status	enum(DRAFT, PENDING, PUBLISHED, REJECTED)	Default `DRAFT`
created_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps, deleted_at	ts	Nullable (SD)
Index	`(tenant_id, subject_id, status)`	

`questions`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
quiz_id	uBigInt	FK → quizzes.id, CASCADE

		CASCADE
type	enum(single_choice, multiple_choice, true_false, short_answer)	NOT NULL
body	text	NOT NULL (معقّم)
explanation	text	Nullable
points	decimal(5,2)	Default 1
sort_order	unsignedInteger	Default 0
source	enum(manual, ai)	Default `manual`
created_by	uBigInt	FK → users.id, Nullable, SET NULL
timestamps, deleted_at	ts	Nullable (SD)
Index	`(tenant_id, quiz_id, sort_order)`	

`question_options`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
question_id	uBigInt	FK → questions.id, CASCADE
body	text	NOT NULL
is_correct	boolean	Default false
sort_order	unsignedInteger	Default 0
timestamps	ts	Nullable

`quiz_attempts`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
quiz_id	uBigInt	FK → quizzes.id, CASCADE
user_id	uBigInt	FK → users.id, CASCADE
attempt_number	unsignedTinyInteger	NOT NULL

status	enum(in_progress, submitted, graded, expired)	Default `in_progress`
score	decimal(6,2)	Nullable
max_score	decimal(6,2)	NOT NULL
passed	boolean	Nullable
started_at	ts	NOT NULL
submitted_at / graded_at	ts	Nullable
timestamps	ts	Nullable
Unique	`(tenant_id, quiz_id, user_id, attempt_number)`	

`quiz_answers`

الحقل	النوع	القيود
id	uBigInt	PK
tenant_id	ulid	FK, CASCADE
quiz_attempt_id	uBigInt	FK → quiz_attempts.id, CASCADE
question_id	uBigInt	FK → questions.id, RESTRICT
selected_option_ids	json	Nullable